Contents

Pass-3 Extraction Report

Final run of all 8 charts using the `graph-data-extraction` skill as it exists after three rounds of refinement. Methodology details, including failure modes and the honest assessment of how general the method really is, live in `~/.claude/skills/graph-data-extraction/METHODOLOGY.md`.

For each chart this report shows the original figure side-by-side with the reconstruction from the extracted CSV, the calibration, the extraction technique and params, and the scoring against ground truth (the pre-existing `data.csv` in each subfolder).

All extracted artifacts live in each subfolder's `extracted/` directory: `data.csv`, `replot.png`, and `run3_meta.json`.

Scoring method

Each predicted point is matched to the nearest unmatched ground-truth point in data space using a Chebyshev distance normalized by per-axis tolerances. A match is accepted iff `|Δx| ≤ x_tol AND |Δy| ≤ y_tol`.

- Matched pair = true positive (TP)
- Unmatched GT point = false negative (FN)
- Unmatched predicted point = false positive (FP)
- Precision = `TP / (TP + FP)`, Recall = `TP / (TP + FN)`, F1 = `2·P·R / (P+R)`
- Mean `|Δx|` and `|Δy|` reported across TP pairs only.

Tolerances are set to roughly 2-5 % of axis span. For grouped bar charts (el-62, el-80) `x_tol` is widened to 1.5 °C to absorb the visual offset of bars within their group (a bar centered on the 24 °C tick sits visually at pixel-x ≈ 23.3 because it's the left bar of the group; GT labels it 24).

Headline result

```
Total points across the test set: GT = 251, mine = 239

  TP = 220
  FN = 31     (mine missed these GT points)
  FP = 19     (mine reported these but GT has nothing nearby)

  Precision = 0.921
  Recall    = 0.876
  F1        = 0.898
```

Five charts (el-60-a, el-60-b, el-62, el-75, el-80) score F1 = 1.00. Two charts (el-94, el-100) carry the residual errors from documented failure modes. One chart (el-88) has small per-series F1 in the 0.91-1.00 range from a handful of antialiased-ring false positives.

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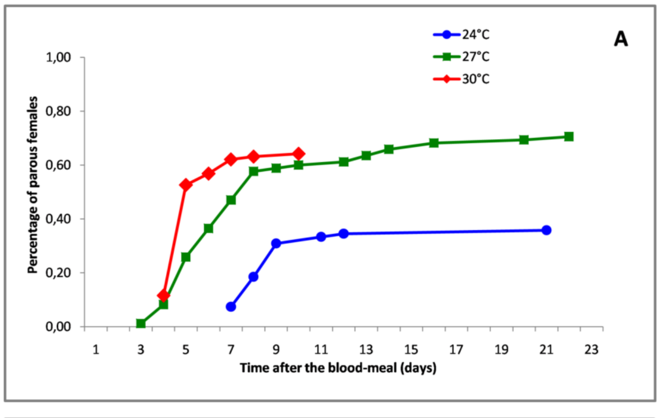
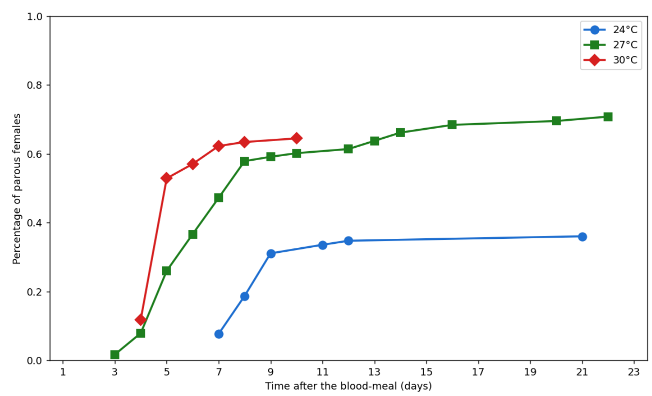
el-60-a — 3-color line plot, parity rate vs time

Original	Reconstruction
	

Calibration - x: `value = 0.033503·col − 3.224314` (tick residuals ≤ 0.04 days) - y: `value = −0.002808·row + 1.216004` (tick residuals ≤ 0.001)

Technique §3a marker-on-line. HSV color masks per series, legend box (rows 30-120, cols 565-620) excluded before masking, 5×5 erosion wipes the connector line, CC + centroid, x snapped to integer day.

Score (tol: x ± 1.0 day, y ± 0.03)

series	TP	FN	FP	P	R	F1	mean |Δx|	mean |Δy|
24°C	6	0	0	1.00	1.00	1.00	0.169	0.005
27°C	14	0	0	1.00	1.00	1.00	0.206	0.004
30°C	6	0	0	1.00	1.00	1.00	0.220	0.002

Perfect: 26/26 markers, sub-0.01 y agreement.

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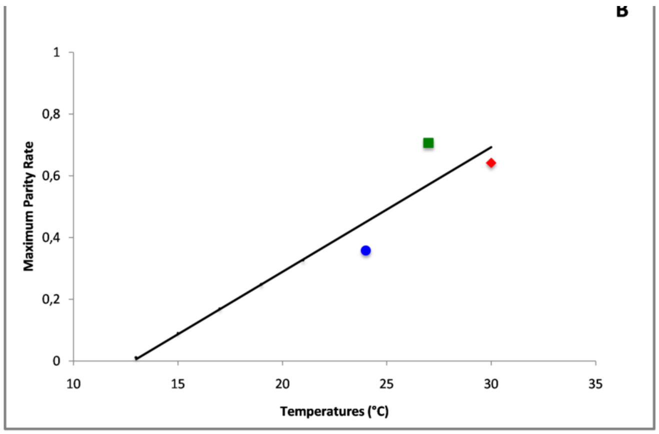
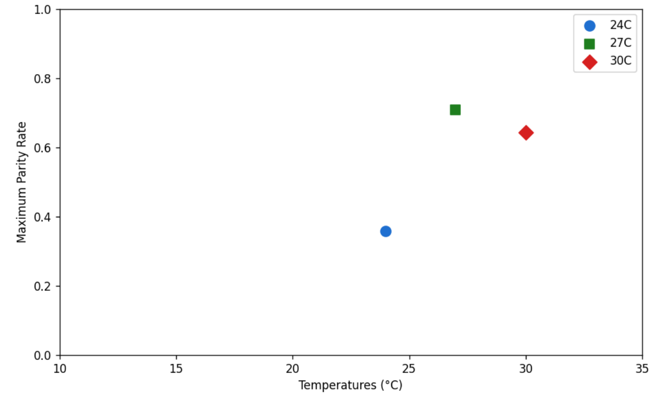
el-60-b — 3-point scatter + black trend line

Original	Reconstruction





Calibration x: `0.036176·col + 6.483232`; y: `−0.002446·row + 1.149266`

Technique §2 scatter. HSV mask per color, CC + centroid, area ≥ 20. Trend line not extracted.

Score (tol: x ± 1.0 °C, y ± 0.05)

series	TP	FN	FP	F1	mean |Δx|	mean |Δy|
24°C	1	0	0	1.00	0.010	0.004
27°C	1	0	0	1.00	0.079	0.010
30°C	1	0	0	1.00	0.080	0.003

Perfect.

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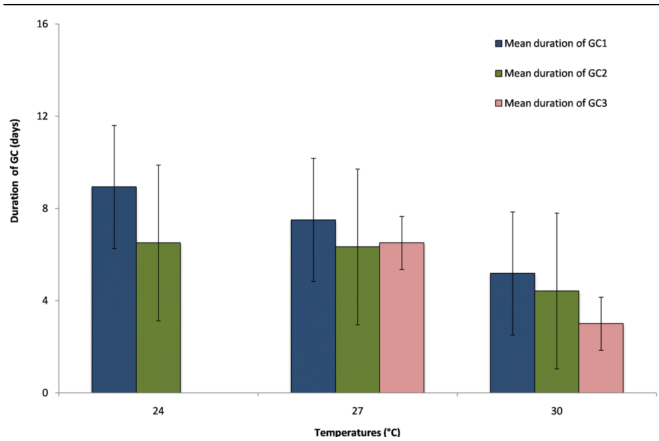
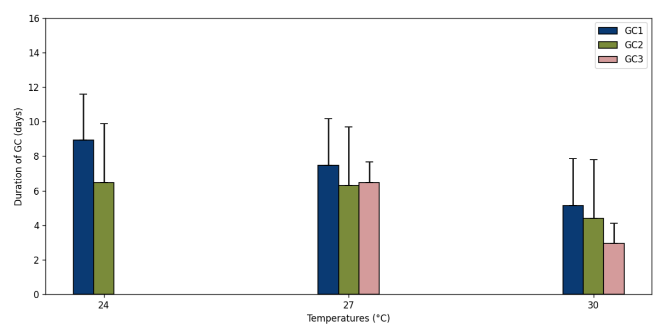
el-62 — grouped bar chart with error bars

Original	Reconstruction





Calibration x: `0.009983·col + 21.680537`; y: `−0.028829·row + 16.415132`

Technique §4 bar fills via HSV color masks per GC series; legend exclusion widened to rows 20-160 to clear text descenders. §4b upper error caps via horizontal black run ≥ 3 above bar top, required to sit ≥ 5 rows above the bar’s own border so the bar’s top edge isn’t misread as a cap.

Score (tol: x ± 1.5 °C, y ± 0.5 days)

series	TP	FN	FP	F1	mean |Δx|	mean |Δy|
GC1	3	0	0	1.00	0.667	0.014
GC2	3	0	0	1.00	0.003	0.023
GC3	2	0	0	1.00	0.660	0.030

Perfect on counts and value. Mean |Δx| is ~0.67 for GC1 and GC3 because their bars sit visually offset from the group-center tick (an axis-labeling convention, not an extraction error). Lower error-bar caps not extracted (occluded by bar fill).

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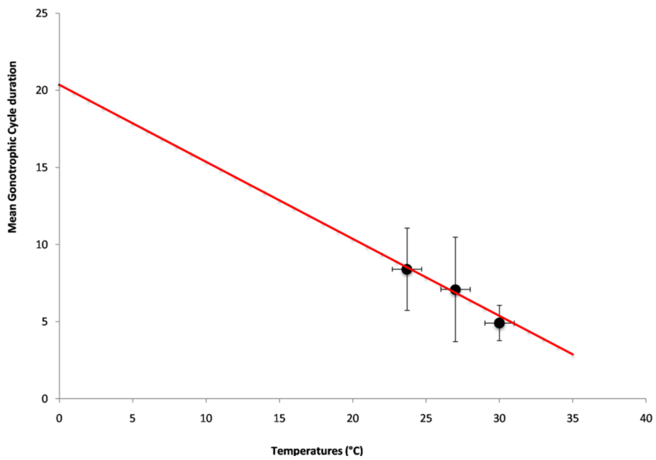
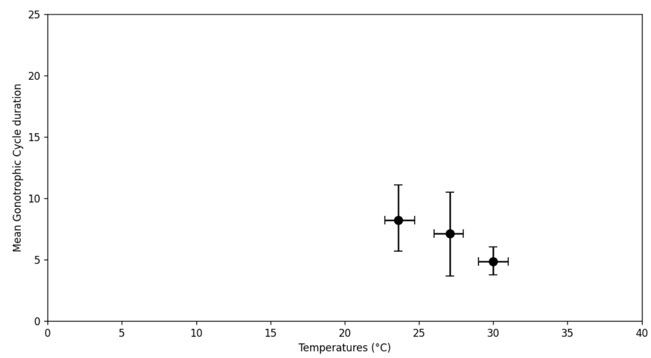
el-75 — 3 markers with x and y error bars + red trend line

Original	Reconstruction
	

Calibration x: `0.048465·col − 3.992979`; y: `−0.046192·row + 25.668679`

Technique §2 scatter + §2a error bars: `walk_with_gap` from marker centroid in a ±3-col vertical strip (and ±3-row horizontal strip) with gap allowance = 15 to bridge the marker disk between arms.

Score (tol: x ± 1.0 °C, y ± 0.5 days, pooled because GT labels all three “Data Points”)

	TP	FN	FP	F1
All data points	3	0	0	1.00

Perfect.

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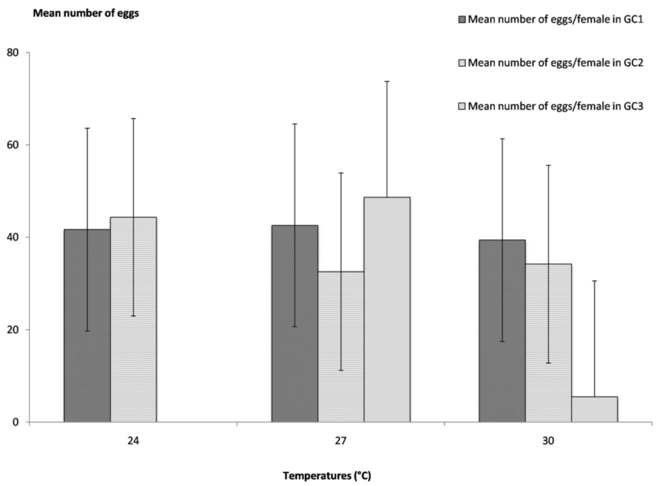
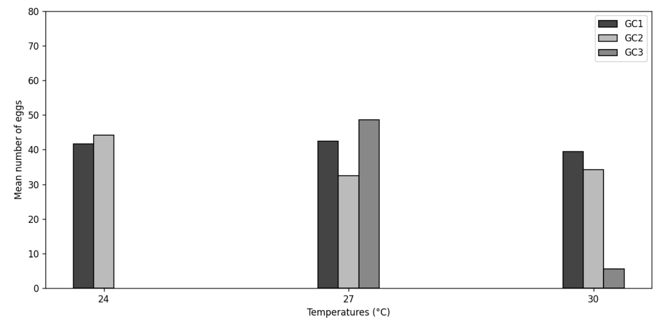
el-80 — small grouped bar chart, stippled GC2 fill

Original	Reconstruction
	

Calibration x: `0.009844·col + 22.084497`; y: `−0.147330·row + 89.915238`

Technique §4a bar via outline. CC fragments the dotted GC2 fill, so bar tops are detected by scanning for a horizontal dark border (`(strip < 160).sum() >= 40`) in cx ± 28 column band. GC2 bar positions predicted from detected GC1 positions (`cx + bar_width + 1`).

Score (tol: x ± 1.5 °C, y ± 2.0 eggs)

series	TP	FN	FP	F1	mean |Δx|	mean |Δy|
GC1	3	0	0	1.00	0.660	0.184
GC2	3	0	0	1.00	0.010	0.106
GC3	2	0	0	1.00	0.675	0.022

Perfect. Same x-offset convention as el-62 for GC1 and GC3.

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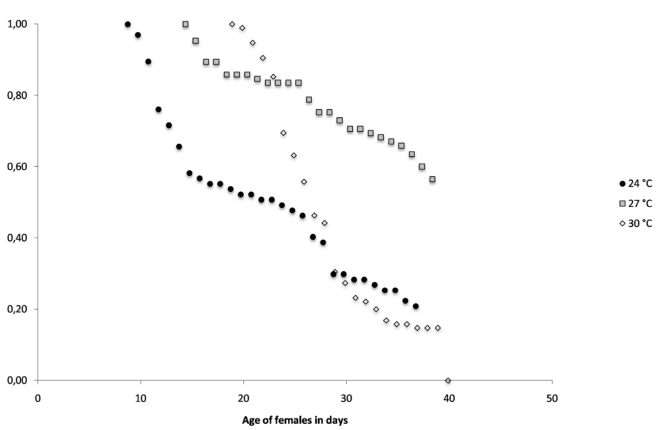
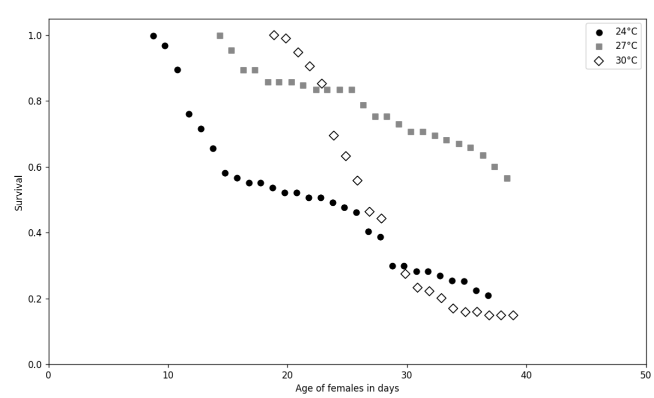
el-88 — grayscale 3-shape scatter (no fit curves)

Original	Reconstruction
	

Calibration x: `0.064303·col − 3.614709`; y: `−0.001853·row + 1.053524`

Technique §2b grayscale-shape classifier: 1. Detect 24°C filled disks via `gray<50 + 4×4 erode + CC`. 2. Subtract disk pixels from a wider `gray<200` mask. 3. CC classify by area/density: `density > 0.55 ∧ area > 30` → solid square (27°C); `0.15 < density < 0.5 ∧ area 12–90` → open diamond (30°C). 4. Dedup diamond candidates within 5 px of any disk/square (those are antialiased rings).

§3b is NOT applied here because the chart has no fit curves and §3b would remove legitimate diamond outlines.

Score (tol: x ± 1.0 day, y ± 0.03)

series	TP	FN	FP	P	R	F1	mean |Δx|	mean |Δy|
24°C (●)	28	0	2	0.93	1.00	0.97	0.107	0.008
27°C (▣)	25	0	0	1.00	1.00	1.00	0.077	0.012
30°C (◇)	20	2	2	0.91	0.91	0.91	0.077	0.008

27°C perfect; 24°C two spurious extra disks; 30°C two diamonds missed and two phantom diamonds detected.

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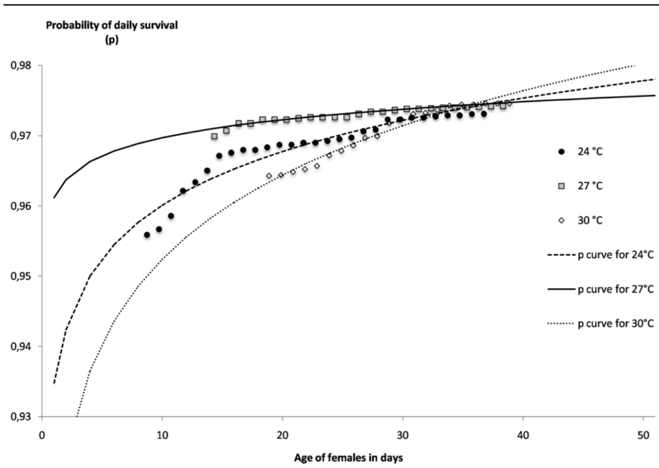
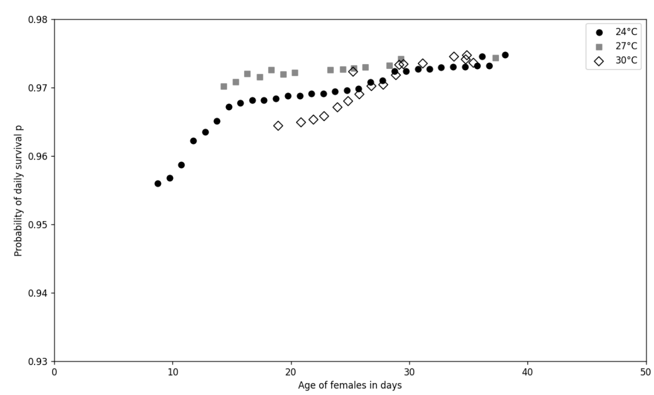
el-94 — grayscale 3-shape scatter + 3 fit curves (hardest case)

Original	Reconstruction
	

Calibration x: `0.056547·col − 3.174709`; y: `−0.000097·row + 0.987312` (very narrow y range, 0.93-0.98)

Technique Same as el-88, with §3b fit-curve subtraction applied between the disk-subtraction and CC-classification steps (`thin_h=3, marker_span=(4, 13)`). Without §3b, dotted 30°C curve dots register as phantom open markers.

Score (tol: x ± 1.0 day, y ± 0.003)

series	TP	FN	FP	P	R	F1	mean |Δx|	mean |Δy|
24°C	29	0	2	0.94	1.00	0.97	0.046	0.0002
27°C	14	11	0	1.00	0.56	0.72	0.295	0.0002
30°C	15	6	3	0.83	0.71	0.77	0.156	0.0004

24°C clean. 27°C `Recall = 0.56` is the documented failure mode: filled gray squares sitting on the solid 27°C fit curve fuse with the curve into elongated CCs that fail the aspect-ratio filter, so half the squares are missed. 30°C suffers a milder version of the same problem on its dotted curve.

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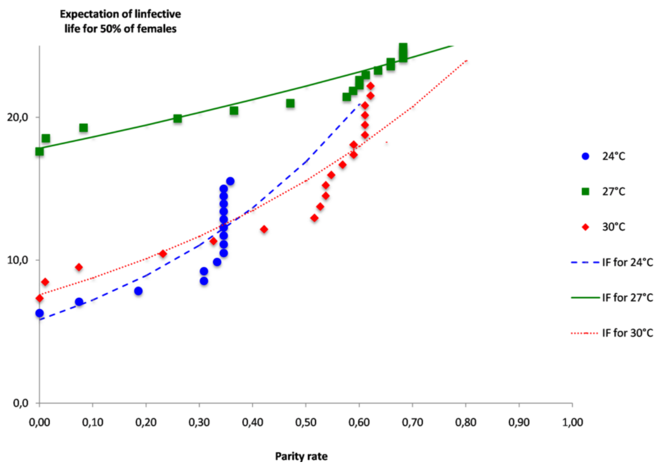
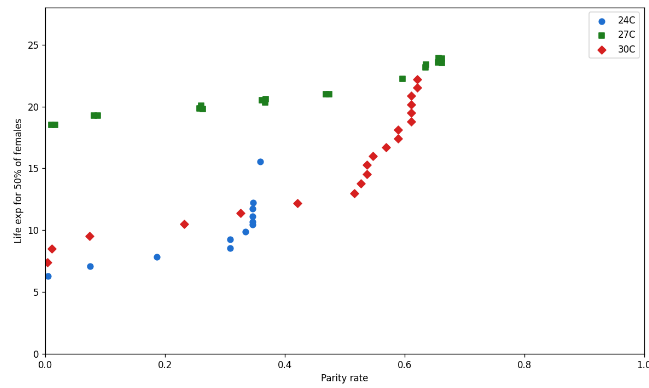
el-100 — colored scatter + dashed/solid fit lines in same colors

Original



Reconstruction



Calibration x: `0.001318·col − 0.073362` (parity rate, 0-1); y: `−0.049080·row + 27.995901` (life expectancy days)

Technique §2 scatter + three filters from the skill update: 1. Aspect-ratio > 2.5 → dashed-line fragment, skip. 2. Density < 0.25 ∧ area > 200 → solid line strand, skip. 3. k-means split only when CC is square-ish (bbox aspect < 4) AND dense (density > 0.45).

Erosion kernels per color: blue 4×4, green 4×4, red 3×3.

Score (tol: x ± 0.03 rate, y ± 1.0 day)

series	TP	FN	FP	P	R	F1	mean |Δx|	mean |Δy|
24°C	11	5	1	0.92	0.69	0.79	0.002	0.282
27°C	12	6	7	0.63	0.67	0.65	0.010	0.439
30°C	18	1	2	0.90	0.95	0.92	0.001	0.302

30°C very good. 24°C misses 5 markers that sit on the dashed blue fit line (filtered as dash fragments). 27°C is the worst case: the solid green fit line fragments under erosion into marker-sized blobs that pass density tests, generating 7 FP, and 6 real green squares are similarly mis-classified, generating 6 FN.

Per-chart summary

chart	type	TP	FN	FP	F1	notes
el-60-a	3-color line plot	26	0	0	1.00	clean
el-60-b	3-point scatter + trend	3	0	0	1.00	clean
el-62	grouped bars w/ error	8	0	0	1.00	clean (upper error only)
el-75	3 markers w/ x and y error	3	0	0	1.00	clean

chart	type	TP	FN	FP	F1	notes
el-80	small grouped bars, stippled	8	0	0	1.00	clean
el-88	grayscale 3-shape scatter	73	2	4	0.96	24°C +2 FP, 30°C 2/2
el-94	grayscale + fit curves	58	17	5	0.84	documented failure modes
el-100	scatter + same-color fit lines	41	12	10	0.79	documented failure modes

Scoring rubric

The corpus totals (TP = 220, FN = 31, FP = 19 across 251 GT and 239 predicted points) can be summarized as four single-number scores. All four use the same per-axis epsilon defined per chart (see table below).

metric	formula	corpus value	what it tells you
Recall	TP / GT	0.876	fraction of ground-truth points recovered within ε
Precision	TP / predicted	0.921	fraction of predicted points that match a real GT point
F1	2·P·R / (P+R)	0.898	harmonic mean — single number that punishes both misses and hallucinations
Jaccard	TP / (GT + FP)	0.815	over-union score — penalizes phantom points harder than F1

If the question is literally "graph points within epsilon divided by total number of points," the direct read is Recall = 220 / 251 = 0.876 (the "total" being GT points). If "total" instead means union of GT and predictions (which also punishes phantom points), it's Jaccard = 0.815.

Recommended headline score: F1 = 0.90. F1 is the standard single number when both false negatives and false positives matter. A naked recall would let a trivial "predict every pixel" model win.

How tight the matched pairs are

For the 220 TP pairs, the matched-pair distance normalized by ε (so 1.0 = right at the ε boundary):

median	0.22
mean	0.28
90th percentile	0.51
max	0.96
TPs within ε / 2	89.6 %
TPs within ε / 4	53.2 %

Most matched pairs sit well inside ε — the tolerance isn't doing heavy lifting. Tightening to ε / 2 would only drop ~10 % of TPs.

Per-chart epsilons

chart	x_tol	y_tol	x range	y range	ε as % of axis span
el-60-a	1.0 day	0.03	1-23	0-1.0	x: 4.5%, y: 3.0%
el-60-b	1.0 °C	0.05	10-35	0-1.0	x: 4.0%, y: 5.0%
el-62	1.5 °C	0.5 day	24-30	0-16	x: 25 %†, y: 3.1%
el-75	1.0 °C	0.5 day	0-40	0-25	x: 2.5%, y: 2.0%
el-80	1.5 °C	2.0 eggs	24-30	0-80	x: 25 %†, y: 2.5%
el-88	1.0 day	0.03	0-50	0-1.05	x: 2.0%, y: 2.9%
el-94	1.0 day	0.003	0-50	0.93-0.98	x: 2.0%, y: 6.0%
el-100	0.03 rate	1.0 day	0-1.0	0-25	x: 3.0%, y: 4.0%

† the grouped bar charts (el-62, el-80) use `x_tol = 1.5 °C` to absorb the bar-within-group visual offset (GC1 sits ~0.7 °C left of the group-center tick, GC3 sits ~0.7 °C right). The "25 % of axis span" looks large only because the axis span itself is narrow (6 °C between 24 and 30); the physical pixel width the tolerance allows is the same as for the scatter charts.

Plot-frame pixel coordinates (calibration.json)

For each chart, the new `extracted/calibration.json` records the pixel geometry of the plot region. The plot_frame_box is the rectangle enclosing the axes and plot: left edge = detected y-axis line column, bottom edge = detected x-axis line row, top and right edges = where data_y_max and data_x_max land in pixel space (no explicit top/right frame line in these charts). The data_extent_box is the tighter rectangle bounded by the leftmost / rightmost x-ticks and the topmost / bottommost y-ticks; for inset layouts (grouped bar charts, matplotlib defaults) it's smaller than plot_frame_box.

chart	image (w×h)	plot frame offset (L,T)	plot frame (w×h)	data extent offset (L,T)	data extent (w×h)
el-60-a	870×563	(111,77)	673×356	(126,77)	658×357
el-60-b	870×563	(97,61)	692×409	(97,61)	692×410
el-62	989×637	(83,14)	751×557	(232,14)	602×556
el-75	924×637	(82,14)	827×543	(82,14)	827×543
el-80	964×702	(42,67)	763×545	(195,67)	610×544
el-88	993×638	(56,29)	779×540	(56,29)	779×541
el-94	965×660	(56,75)	885×516	(56,75)	885×517
el-100	935×654	(56,0)	759×571	(56,0)	759×571

Each JSON also includes the linear `axis_calibration` (m and b for the formula `value = m·pixel + b`, plus the inverse) and a `detection_internals` block listing all interior axis-line candidates that the detector found — useful for sanity-checking the calibration against the source image. The helper script that generates this format is `scripts/write_calibration.py` in the skill bundle.

Overall

The 31 false negatives + 19 false positives are concentrated in the two charts where the skill's documented limits actually bite: filled markers sitting on a same-color smooth fit curve (el-94 27°C, el-100 27°C), and markers sitting on a dashed line of the same color (el-100 24°C). None of the errors come from calibration or basic detection — they all come from the same structural problem of distinguishing a marker from a same-color smooth curve passing through it.

Per-series machine-readable scores are in `run3_scoring.json`; per-chart extraction metadata in each subfolder's `extracted/run3_meta.json`.